

Q-77C VLAD

- Microprocessor-Controlled Full-Function Selector
 - Electronic Function Select before launch
 - Command Function Select after launch
- Multi-channel RF Capability
- Multiple Depth and Life Selection Capability
- Convergence Zone Receive Beam
- Launchable from Existing ASW Platforms



The AN/SSQ-77C VLAD (Vertical Line Array DIFAR) sonobuoy is a passive, tactical search and surveillance sonobuoy designed to improve detection and tracking capability for the DIFAR system in a noisy, high traffic environment.

The VLAD concept utilizes a vertical line array of omnidirectional hydrophones in place of the single hydrophone used in the standard DIFAR unit. A directional hydrophone is mounted at the array phase center and provides target bearing data.

The signal format for the VLAD sonobuoy is identical to that of the AN/SSQ-53F (DIFAR) and compatible with the AN/AQA-7 processor. All beam-forming functions are accomplished with the sonobuoy, with provisions for sea noise equalization and omnidirectional phase tracking.

The VLAD may be dropped from an aircraft at airspeeds to 370 knots and from altitudes to 30,000 feet. Descent is stabilized and slowed by a parachute.

Operating parameters are programmed and verified by means of push-buttons and a pair of seven-segment LED displays located on the side of the unit. These parameters are stored and maintained in the buoy's non-volatile memory, and are implemented upon activation of the sea water battery.

SPARTON

Passive Sonobuoy

Sparton Government Business Division

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TypePassive, DIFAR with VLA omni
SizeA
Weight.....23 pounds (10.4 kilograms)
Launch Altitude40 to 30,000 feet (12 to 9144 meters)
Launch Speed.....0 to 370 KIAS
Operational Depth.....200, 500, and 1000 feet (61, 152, and 305 meters)
Operational Life0.5, 1.0, 2.0, 4.0, and 8.0 Hours (EFS or CFS
selectable)
RF Channels.....96 (136 to 173.5 MHz) (EFS or CFS selectable)
RF Power.....1.0 watt minimum
Scuttle8 hours (automatic)
Audio Frequencies.....10 to 2400 Hz

Technical Specifications